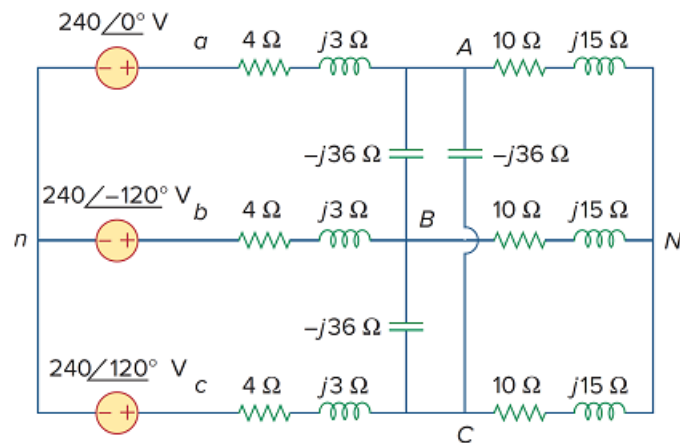


Problem

Given the circuit in Fig. 12.67, use *PSpice* or *MultiSim* to determine currents I_{aA} and voltage V_{BN} .

Figure 12.67



Step-by-step solution

Step 1 of 2

The schematics is shown. Pseudo components IPRINT & PRINT are inserted to measure, I_{aA} , V_{BN} . In the AC sweep box, we set Total pts = 1, Start frequency = 0.1892, End frequency = 0.1892. Once the circuit is simulated, we get the output file as,

<i>Frequency</i>	<i>VM(2)</i>	<i>VP(2)</i>
$1.592E-01$	$2.308E+02$	$-1.334E+02$
<i>Frequency</i>	<i>1M(V_PRINT2)</i>	<i>1P(V_PRINT2)</i>
$1.592E-01$	$1.15E+01$	$3.699E+01$

Step 2 of 2

From which,

$$I_{aA} = 11.15 \angle 37^\circ \text{ A}$$

$$V_{BN} = 230.8 \angle -133.4^\circ \text{ V}$$